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MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 4.1

Vector Spaces and Subspaces

Vector Spaces

A vector space is a nonempty set V of objects (vectors) on which are defined two operations (addition and scalar multiplication) subject to the following ten axioms.

The axioms must hold for all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V and for all scalars c and d .

Scalars can come from $\mathbb{R}$ or $\mathbb{C}$, but unless otherwise indicated, we assume in this chapter that scalars are real numbers.

Vector Space Axioms

- The sum of $\mathbf{u}$ and $\mathbf{v}$, denoted by $\mathbf{u} + \mathbf{v}$, is in V .
 V is closed with respect to vector addition.
- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
Vector addition is commutative.
- $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
Vector addition is associative.
- There is a zero vector $\mathbf{0}$ in V such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
 $\mathbf{0}$ is called the additive identity.
- For each $\mathbf{u}$ in V , there is a vector $-\mathbf{u}$ in V such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.
 $-\mathbf{u}$ is called the additive inverse of $\mathbf{u}$.

Vector Space Axioms

6. The scalar multiple of $\mathbf{u}$ by c , denoted by $c\mathbf{u}$, is in V .
 V is closed with respect to scalar multiplication.
7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
Distributive property.
8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
Distributive property.
9. $c(d\mathbf{u}) = (cd)\mathbf{u}$
Scalar multiplication is associative.
10. $1\mathbf{u} = \mathbf{u}$
 1 is called the multiplicative identity.

Examples of Vector Spaces

$$\mathbb{R}^n, n \geq 1$$

$$\mathbb{P}_n, n \geq 0$$

$\mathbb{P}_n$ is the set of polynomials of degree at most n .

Example 1

Let V be the first quadrant in the xy -plane.

$$V = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x \geq 0, y \geq 0 \right\}$$

- a) If $\mathbf{u}$ and $\mathbf{v}$ are in V , is $\mathbf{u} + \mathbf{v}$ in V ?
- b) If $\mathbf{u}$ is in V and c is a scalar, is $c\mathbf{u}$ in V ?

Example 2

Show that a line which does not pass through the origin in $\mathbb{R}^2$ is not a vector space.

Is a line through the origin in $\mathbb{R}^2$ a vector space?

Subspaces

A subspace of a vector space V is a subset H of V that has the following three properties.

- The zero vector of V is in H .
- H is closed under vector addition.
That is, for all $\mathbf{u}$ and $\mathbf{v}$ in H , $\mathbf{u} + \mathbf{v}$ is in H .
- H is closed under scalar multiplication.
That is, for all $\mathbf{u}$ in H and all scalars c , $c\mathbf{u}$ is in H .

Subspaces

Since all elements in H are also in V , all of the remaining vector space axioms are also satisfied. Thus, H is itself a vector space!

It is generally much easier to prove that a given set forms a vector space by showing it is a subspace of a known vector space than by starting from scratch.

Example 2 (continued)

Show that a line through the origin in $\mathbb{R}^2$ is a subspace of $\mathbb{R}^2$ (and is thus a vector space).

Example 3

Show that $\mathbb{R}^2$ is not a subspace of $\mathbb{R}^3$.

Then, show that

$$H = \left\{ \begin{bmatrix} s \\ t \\ 0 \end{bmatrix} : s, t \text{ real} \right\}$$

is a subspace of $\mathbb{R}^3$.

Example 4

Determine whether the set is a subspace of $\mathbb{P}_n$ for an appropriate value of n .

All polynomials of the form $\mathbf{p}(t) = a + t^2$, where a is in $\mathbb{R}$.

Example 5

Given two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in a vector space V , show that $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ is a subspace of V .

Subspaces and Span

If $\mathbf{v}_1, \dots, \mathbf{v}_p$ are in a vector space V , then $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is a subspace of V .

$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is the subspace spanned by $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.

Given any subspace H of V , a spanning set for H is a set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ such that $H = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.

Example 6

Let H be the set of all vectors of the form $\begin{bmatrix} 2t \\ 0 \\ -t \end{bmatrix}$.

Show that H is a subspace of $\mathbb{R}^3$.

Example 7

Let W be the set of all vectors of the form shown, where a , b , and c represent arbitrary real numbers. In each case, determine whether or not W is a vector space.

$$\begin{bmatrix} 3a + b \\ 4 \\ a - 5b \end{bmatrix}$$

$$\begin{bmatrix} 4a + 3b \\ 0 \\ a + b + c \\ c - 2a \end{bmatrix}$$

Example 8

Prove that the zero vector in a vector space is unique.

Example 9

For what value(s) of h will $\mathbf{y}$ be in the subspace of $\mathbb{R}^3$ spanned by $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$?

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$$
